

Homework 7

Problem 1

Xavier and Yolanda are playing roulette. They both place bets on red on the same 3 spins of the roulette wheel before Xavier has to leave. (Remember, the probability that any bet on red on a single spin wins is $18/38$.) After Xavier leaves, Yolanda places bets on red on 2 more spins of the wheel. Let X be the number of bets that Xavier wins and let Y be the number that Yolanda wins.

1. Compute and interpret $P(X + Y = 4)$.
2. Make a table representing the marginal distribution of $X + Y$.
3. Use the table from the previous part to compute $E(X + Y)$, and verify that it is equal to $E(X) + E(Y)$. (You computed $E(X)$ and $E(Y)$ in a previous homework.)
4. Without doing any computation, determine if $\text{Var}(X + Y)$ will be greater than, less than, or equal to $\text{Var}(X) + \text{Var}(Y)$. Explain your reasoning.
5. Use the marginal distribution of $X + Y$ (from earlier in this problem) to compute $\text{Var}(X + Y)$.
6. In a previous HW you identified the marginal distribution of X by name. Use facts about that distribution to compute $\text{Var}(X)$.
7. In a previous HW you identified the marginal distribution of Y by name. Use facts about that distribution to compute $\text{Var}(Y)$.
8. Verify that $\text{Var}(X + Y)$ is equal to $\text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$. (You computed the covariance in a previous homework.)

x, y	0	1	2	3	4	5
0	0.0404	0.0727	0.0327	0	0	0
1	0	0.1090	0.1963	0.0883	0	0
2	0	0	0.0981	0.1766	0.0795	0
3	0	0	0	0.0294	0.0530	0.0238

Problem 2

A vendor at a weekend farmers market sells three products: jars of honey, loaves of sourdough bread, and bags of ice. Let X be the number of jars of honey sold, Y the number of loaves sold, and Z the number of bags of ice sold. Based on several seasons of sales data, the number of units sold on a given Saturday follows a joint distribution with

- Mean 40 and standard deviation 8 for honey jars sold,
- Mean 25 and standard deviation 5 for bread loaves sold,
- Mean 15 and standard deviation 6 for ice bags sold,
- Correlation of 0.5 between honey jars sold and bread loaves sold,
- Correlation of -0.4 between honey jars sold and ice bags sold. (Cooler days tend to bring out more shoppers browsing specialty foods like honey, but fewer people buying ice.)

Honey jars sell for \$12 each, bread loaves sell for \$7 each, and ice bags sell for \$4 each.

1. Let $R_1 = 12X + 7Y$ be the vendor's revenue from honey and bread combined. Compute $E(R_1)$ and $SD(R_1)$.
2. Let $R_2 = 12X + 4Z$ be the vendor's revenue from honey and ice combined. Compute $E(R_2)$ and $SD(R_2)$.
3. Without doing any additional calculation, explain whether you'd expect each of $SD(R_1)$ and $SD(R_2)$ to be larger or smaller if the variables were uncorrelated.

Problem 3

Devi and Paxton are meeting. Arrival times are measured in minutes after noon, with negative times representing arrivals before noon. Devi's arrival time follows a Normal distribution with mean 20 and SD 15 minutes, and Paxton's arrival time follows a Normal distribution with mean 25 and SD 10 minutes.

All of the probability calculations will involve a Normal distribution, but make sure you clearly identify the correct mean and SD. You should also compute standardized values and use a table to make a ballpark estimate of probabilities before jumping straight to software/calculator.

You will compute all of the following probabilities in each of the 3 scenarios below.

- a. The probability that Devi arrives before noon.
- b. The probability that Devi arrives before Paxton.
- c. The probability that Devi arrives before Paxton given that Paxton arrives at 12:10.
- d. The probability that the first person to arrive has to wait more than 15 minutes for the second person to arrive.

1. Scenario 1: Assume the arrival times are independent
2. Scenario 2: Assume the pairs of arrival times follow a Bivariate Normal distribution with correlation 0.8
3. Scenario 3: Assume the pairs of arrival times follow a Bivariate Normal distribution with correlation -0.7

Problem 4

Consider the general problem of estimating a population proportion p for some binary (success/failure) categorical variable of interest. Select a sample of size n and let X be the number of successes in the sample.

1. We are considering a general problem here, but describe a specific context where you might want to do this: what is the population and its rough size, what is the variable of interest and what is “success”, what is the sample, how do you measure X for the sample? (For example, the handouts covered the general problem of estimating μ for a Poisson population, but we thought of it in terms of the car dealership problem.)
2. Suggest an estimator of p ; it should be an expression involving X and n . Describe in words what your estimator is. (Hint: what is the most obvious estimator you can think of?)
3. Suppose you observe $x = 6$ successes in a sample of size $n = 10$. Use your estimator to compute a point estimate of p based on this sample.
4. Consider the estimator (called the “plus four” estimator)

$$\frac{X+2}{n+4} = \left(\frac{4}{n+4}\right)(0.5) + \left(\frac{n}{n+4}\right)\left(\frac{X}{n}\right)$$

Explain in words what this estimator does. (The two expressions in the equation above are just different ways of expressing the estimator.)

5. Suppose you observe $x = 6$ successes in a sample of size $n = 10$. Use the plus four estimator to compute a point estimate of p based on this sample.
6. If you observe $x = 6$ successes in a sample of size $n = 10$, which of your two estimates is a better estimate of p ? Explain.